INDIVIDUAL MILITARY SPECIFICATION SHEET
ELECTRON TUBE, MAGNETRON, PULSE
TYPE 7521

Applicable paragraphs of Military Specification MIL-E-1, as referenced herein, from a part of the Specification Sheet (Note 20)

DESCRIPTION: Magnetron, Pulsed type, mechanically tunable 8900 to 9400 MC., .80kw Peak Power Output (min.), integral magnet, convection and conduction air cooled, unipotential cathode.

ABSOLUTE-MAXIMUM RATINGS (SEE NOTE 1):

INDEPENDENT: (Note 1a):

Parameter	Ef	eb	tk	VSWR	T(anode)	Tuner Torque
Units	V	kv	sec	-	°C	in.-oz.
Maximum	5.5	5.0	-	1.5	100	60
Minimum	4.5 (see notes 2 and 3)	-	60	- (see note 3)	- (see note 5)	- (see note 4)

DEPENDENT: (Note 1b):

Parameter	Ib	ib	Pi	pi	Du	tpc
Units	mAdc	a	W	kw	-	us.
Maximum	2.0	1.05	7.5	5.0	.002	1.5
Minimum	-	-.5	-	-	-	-

STORAGE, HANDLING, AND INSTALLATION:

Cooling: Convection and conduction
Magnet isolation: See note 6
Mounting support: See Figure 1
Mounting position: Any
Tuner Drive mechanism: See note 13
Preparation for shipment: See note 23
Output coupling: UG-39/U: See figure 1
Input connections: UG-290A/U: See figure 1
Vibration, shock: See notes 10, 11
Weight: Approx 1.5 lbs.

Par. No.	Test	Conditions	Symbol	Limits Min	Limits Max	Units
3.1	Qualification	Required (Note 21, 22)	-	-	-	-
3.6	Performance		-	-	-	-
4.5	Holding period	t = 168 hr.	-	-	-	-
4.9.2	Dimensions	(figure 1)	-	-	-	-

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Ref	Test	Conditions	Symbol	Min	Max	Units
4.9.7	Moisture-Vaporproof Barrier pack					
4.9.18	Carton Drop	Package Group 9, Container Size A, Drop Test 1				
4.9.19.1	*Vibration	No Voltages, Note 10, F = 25				
4.9.19.2	**Vibration	No Voltage, Note 10, F = 55				
-	**Shock Test	Note 11; G = 15				
-	*Tuner Drive Torque	T = 25°C(Approx.)	Torque:	10	55	in. oz.
-	**Tuner Drive Torque	T = 55°C to 85°C (Approx.) Note 12	Torque:	10	55	in. oz.
4.9.17	**Mechanical Fatigue	T = 25°C (Approx.) Note 16		2500	-	cycles
4.10.8	*Heater Current	Ef = 5.0 V tk = 60 sec. (min.)	If:	.40	.50	A
4.16.8	Oscillation (1)					
-	Standing Wave Ratio	VSWR = 1.1 max. except as otherwise noted				
4.16.3.2	Heater Cathode Warmup Time	tk = 60 sec max at Ef = 5.0 V Ef = 5.0 V for test Note 8				
4.16.3.3	Pulse Characteristics	tpc = 0.2 ± 0.02 us; Du = .001 trv = 0.09 to 0.11 us, Note 2; 7.				
4.16.3.4	Anode Current	ib = .90 a				
4.16.3.5	Pulse Voltage	F1, F2, F3	epy:	4.0	4.5	kv
-	Power Output	F1 = 8900 Mc F2 = 9150 Mc F3 = 9400 Mc VSWR = 1.5 Min. Power Phase	Po:	.80	-	W

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Ref	Test	Conditions	Symbol	Min	Max	Units
4.16.3.7	R.F. Bandwidth	F1, F2, F3; VSWR = 1.5 Worst Phase ib = .85 to 1.05 a	BW	-	2.2/ tpc	Mc
4.16.5	Pulling Factor	F1, F2, F3	ΔF:	-	20	Mc
4.16.6	*Pushing Factor	F1, F2, F3 Note 17	ΔF:	-	3.5	Mc
4.10.7.3.2	Tunable Frequency		F:	8900	9400	Mc
4.10.7.3.2	**Tunable Frequency	Note 13	F:	8900	9400	Mc
-	Stability Test	Note 14 VSWR = 1.5 min. F1, F2, F3	M.P.:		0.25	%
4.9.14	**Temp. Coeff.	Anode T, Note 5 T = 25°C to 100°C; F1, F3	ΔF/°C:	-	.20	mc/°C
4.9.15.1	**Low Temperature Operation	F3; -65°C Ambient tk = 60 sec. (max.)	M.P.:	-	0.5	%
4.9.15.2	**High Temperature Operation	+75°C Ambient F3				
-	**Anode-Cathode Capacity		C:	-	10	uuf
	Shelf Life Stability	t = 30 days, VSWR = 1.5 Note 18, 14 tk = 60 max.	M.P.:	-	.5	%
4.11.5	Intermittent Life Test	Osc. (1); F2, Group D; Note 15; VSWR = 1.5° min. cycles through λ G in about 30 mins.	t	1500	-	cycles
4.11.4	Life Test End Point	Note 19				
	R.F. Bandwidth		BW:	-	3.0/ tpc	Mc
	Pulse Voltage		epy:	-	4.9	kv
	Power Output		Po:	0.680	-	W
	Stability		M.P.:	-	0.5	%

NOTE 1: For the assistance of designers of electronic equipment, the absolute ratings have been divided into two groups as follows:

a. Independent: These ratings shall not be exceeded. They are limiting values beyond which the serviceability of any individual tubes may be impaired.

b. Dependent: These ratings are interrelated, but it does not necessarily follow that combinations of limits can be attained simultaneously.

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NOTE 2: The characteristics of the applied pulse must be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. The pulse characteristics listed under Oscillation 1, (4.16.3.3), are those obtained with modulators used in manufacturing processes and are not necessarily applicable with field type modulators.

NOTE 3: Frequency skipping or unstable operation may be encountered at some phase position when the mismatch occurs at the end of a "long" line, (two meters or longer).

NOTE 4: The maximum tuner torque specified cannot be exceeded without causing permanent damage to the tube.

NOTE 5: The temperature shall be measured at the point indicated on the Electron Tube Drawing. Temperature coefficient measurements shall be taken at 25 degree intervals across the indicated temperature range. The normal anode temperature during operation is 20°C greater than ambient temperature. At no time shall frequency change more than the specified temperature coefficient limit.

NOTE 6: In handling and mounting the magnetron, care must be exercised to prevent demagnetization. Ferromagnetic materials or energized magnets shall not be brought within two inches of the tube.

NOTE 7: The pulse characteristics of Oscillation (1) shall be as follows: No spike or ripple shall exceed ± 7% of the average peak value of voltage or current. The current pulse fall time shall not exceed 0.2 us (max.) as measured between 0 and 85 percent levels. Pulse shall be applied directly to plates of oscilloscope.

NOTE 8: See Electron Tube Drawing. Heater-cathode power is applied to the BNC type connector as specified on the outline drawing. The center conductor is the cathode connector.

NOTE 9: Reasonable care should be used in the storage, installation, and use of the tube to avoid imparting vibration or shock in excess of the values for which it is designed to withstand.

NOTE 10: The tube shall be mounted on a test plate and vibrated along each of three (3) mutually perpendicular axes at the specified frequency at .08 inches total excursion for 60 seconds. The tube shall meet the requirements of Osc. 1 after vibration.

NOTE 11: The magnetrons shall be mounted on a test plate and dropped 10 times on each of three (3) mutually perpendicular axes parallel to the reference planes shown on the Electron Tube Drawing. The shock pulse shall have a duration of approximately 11 milliseconds as measured at the quarter amplitude points of the acceleration shock wave. The tube shall meet the requirements of osc. 1 after this shock.

NOTE 12: The tuner drive mechanism shall not be "set" against either mechanical stop.

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NOTE 13: The tube shall be capable of being mechanically tuned from 8900 to 9400 megacycles by turning the tuner shaft 1.68 turns ± 17% with the anode current held constant.

NOTE 14: Stability shall be measured with the VSWR adjusted to that phase which produces maximum instability. The missing pulses, (M.P.) shall be counted during the last three minutes of a test interval not to exceed six minutes, and shall not exceed the limit specified. A missing pulse is defined as one whose energy is less than 70% of a normal pulse, within ± 1 percent frequency range of the normal test frequency.

NOTE 15: Life Test shall be conducted in accordance with the following operational cycle:

Condition	Duration	Ef	Ib
(1)	5 min	0V	0
(2)	1 min	5.0V	0
(3)	12 min	5.0V	.90a

The above cycle shall be repeated until the accrued time of condition (3) is 300 hours; the tube shall meet end of life requirements at that time.

NOTE 16: A tuning cycle consists of two complete excursions each in opposite directions through the tuning range of the magnetron. The tube shall meet end of life requirements at completion of this test.

NOTE 17: The Anode Current shall be varied between .85 and 1.0 peak amperes at a rate sufficient to prevent thermal effects.

NOTE 18: This test shall be performed on four tubes per month when the tube is in continuous production, but shipments of that month's production shall not be held pending completion of this test. So long as three of the four tubes for each of the first three months of a production run pass the test and 75% of the cumulative quantity of tubes tested pass the test, tubes shall be considered to conform to this specification. If either of the conditions are not met, shipment shall be halted until three of four tubes of current production conform to the test.

NOTE 19: The tube shall pass all applicable production tests at the end of the specified life, with the condition that the criteria for acceptance be modified in accordance with the life test end points listed.

NOTE 20: Referenced documents shall be of the issue in effect on the date of invitation for bids.

NOTE 21: Application for Qualification Inspection should be made in accordance with Appendix A of Military Specification MIL-E-1, with the exception that the request for Qualification Inspection shall be addressed to Components and Materials Branch, Standardization Division, U. S. Army Signal Equipment Support Agency, Fort Monmouth, New Jersey.

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NOTE 22: Magnetrons which conform to all requirements of this specification and have passed Government inspection, procured directly or indirectly on Army contracts, shall be marked "USA" either preceding or above the qualification code and type designation as specified in the bid request or contract. The letters "JAN" or any abbreviation therefore shall not be used. If any specification variance or waiver has been granted, only the commercial brand designation shall be used.

NOTE 23: Tubes shall be prepared for domestic or overseas shipment, as specified in the contract or order, in accordance with Specification MIL-E-75/1. (See Note 20.)

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patent invention that may in any way be related thereto.

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